ANDROID PROGRAMMING PRACTICAL 12

Module 2 · Fetch and parse JSON from a web service

Aim

Fetch and parse JSON data from a web service.

Project name: JSONDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create an activity with a button and a TextView or RecyclerView for results.

2. Add the INTERNET permission to AndroidManifest.xml.

3. Choose a reachable test API and inspect its JSON structure before writing the parser.

4. Perform network access off the main UI thread using an executor, thread, or suitable networking library.

5. Parse the response with JSONObject/JSONArray or a JSON library, matching the actual response schema.

6. Update the UI on the main thread and handle HTTP errors, malformed JSON, and no connection.

7. Run the app with internet access and verify parsed fields are displayed; test failure handling.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

The app fetches a JSON response and displays parsed fields, or shows a clear error on failure.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.